

Generation Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q17-0814

Ornl 138 kV
April 17, 2019

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q17-0814. The Interconnection Customer proposes a 59.3 MW solar photovoltaic facility interconnecting at the Ornl 138 kV substation in the Maryland area. The request was received on June 26, 2017. The study evaluated system intact and single contingency conditions using the regional planning case, including 263 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 2 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$322,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: FREDERICK RD – NINE SPRINGS 138 KV L
Contingency: SIERRA PACIFIC LUMBER 138/1 kV Transformer
Loading: 103.8% (Rating: 270 MVA)
DFAX: 6.5%
Pre-project loading: 102.3%
2. Facility: NINE SPRINGS – VIADUCT AVENUE 138 KV line
Contingency: SIERRA PACIFIC LUMBER 138-1 kV Transformer
Loading: 102.2% (Rating: 220 MVA)
DFAX: 6.5%
Pre-project loading: 100.5%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
FREDERICK RD – NINE SPRINGS 138 KV L	Reconductor line	\$948,000	13% (\$125,000)
NINE SPRINGS – VIADUCT AVENUE 138 KV	Reconductor line	\$1,499,000	13% (\$197,000)
	Total allocated to project		\$322,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.